WORKSHEET

Chapter 2: Complex number operations

1 Simplify each of the following expressions:

a $3(5 + i) + 4(3 - i)$

b $5(3 - 2i) + i(6 - 3i)$

c $-2(3 - 2i) - i(6 - 3i)$

d $2(9 + 2i) + 3(11 - 2i)$

e $4(-5 + 7i) - 2i(6 + 2i)$

f $3i(3 - 4i) - (7 - 4i)$

© Cengage Learning Australia Pty Ltd 2014

2 Simplify each of the following expressions:

a $(4 + i)(7 - i)$

b $(5 - 2i)(2 - 7i)$

c $(8 - 3i)^2$

d $\left(\sqrt{3} + 2i\right)\left(\sqrt{3} - 2i\right)$

e $(-6 + 5i)^2$

f $(8 - 3i)(5 - 6i)$

3 Find the values of u and v, where u and v are real, if:

a $u + vi = 3(4 - 2i) + i(3 + i)$

b $u + vi = 4(2 + 3i) + 2i(3 - 5i)$

© Cengage Learning Australia Pty Ltd 2014

c $u + vi = 5(1 - 7i) - 3i(2 + 3i)$

d $u + vi = 2i(8 + i) + 5i(2 - 3i)$

e $u + vi = 3i(2 - 5i) + 6i(4 - 7i)$

f $u + vi = 3i(7 - 2i) - 3i(4 - 9i)$

4 If $a = 6 + 5i$ and $b = 3 - 7i$ find:

a $a + 3b$

b $3a - 5b$

c ab

d $\frac{a}{b}$

e $\frac{3a-4}{4b+3i}$

f $(a + 5i)(4b - 2)$

5 Realise the denominator for each of the following complex fractions:

a $\frac{1}{2 + i}$

b $\frac{1 - 3i}{5 + 3i}$

c $\frac{\sqrt{11} + 3i}{\sqrt{11} - 3i}$

© Cengage Learning Australia Pty Ltd 2014

d $\dfrac{2-3i}{5+7i}$

e $\dfrac{4-5i}{8-i}$

f $\dfrac{2\sqrt{3}-5i}{\sqrt{3}+4i}$

6 If $z = 3 - i$ and $w = -2 + 5i$ find:

a zw

b $(z+i)(w-2)$

c $\overline{z-w}$

d $\overline{z+w}$

e $\overline{z} \times w^2$

f $(z+i)(\overline{w}-2)$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $27 - i$

b $18 - 4i$

c $-9 - 2i$

d $51 - 2i$

e $-16 + 16i$

f $5 + 13i$

2 **a** $29 + 3i$

b $-4 - 39i$

c $55 - 48i$

d 7

e $11 - 60i$

f $22 - 63i$

3 **a** $u = 11, v = -3$

b $u = 18, v = 18$

c $u = 14, v = -41$

d $u = 13, v = 26$

e $u = 57\ v = 30$

f $u = -21, v = 9$

4 **a** $15 - 16i$

b $3 + 50i$

c $53 - 27i$

d $\frac{-17}{58} + \frac{57}{58}i$

e $\frac{-207}{769} + \frac{530}{769}i$

f $340 - 68i$

5 **a** $\frac{2}{5} - \frac{1}{5}i$

b $\frac{-2}{17} - \frac{9}{17}i$

c $\frac{1}{10} + \frac{3\sqrt{11}}{10}i$

d $\frac{-11}{74} - \frac{29}{74}i$

e $\frac{37}{65} - \frac{36}{65}i$

f $\frac{-14}{19} - \frac{13\sqrt{3}}{19}i$

6 **a** $-1 + 17i$

b $-12 + 15i$

c $5 + 6i$

d $1 - 4i$

e $43 - 81i$

f $-12 - 15i$

© Cengage Learning Australia Pty Ltd 2014